

Certainly! Below is a simple Python implementation of the **Bubble Sort** algorithm, which you can run locally with Python

```
def bubble_sort(arr):
    n = len(arr)
    # Traverse through all array elements
    for i in range(n):
        # Last i elements are already in place
        for j in range(0, n - i - 1):
            # Swap if the element found is greater than the next element
            if arr[j] > arr[j + 1]:
                arr[j], arr[j + 1] = arr[j + 1], arr[j]
    return arr

# Example array
array = [1, 5, 6, 7, 8, 9, 10, 4, 3]
# Sorting the array
sorted_array = bubble_sort(array)
print("Sorted array:", sorted_array)
```

Explanation:

1. **bubble_sort(arr)** iterates over the list multiple times.
2. In each pass, adjacent elements are compared and swapped if they are out of order.
3. After each full pass, the largest unsorted element ?bubbles up? to its correct position.
4. The process repeats until no swaps are needed, indicating the list is sorted.

Running this code will output:

CODE

Sorted array: [1, 3, 4, 5, 6, 7, 8, 9, 10]

Feel free to modify the array or try other sorting algorithms (e.g., Insertion Sort, Selection Sort) if you'd like! ?